ACCOUNTING & BALANCING

1. Financial Accounting as Information Source

Definition- Process vs. System
Decision Making r
4 Objectives
Understanding Story/ Business & Investment Decisions

1.1. *Business Activities 3 & Role of Accounting*

Knowledgeable RUs of accounting info *Wider BE* Simon 2020

- 3 Different Activities- list & define
    - Define Shareholders
- CHART of business activities
- 3 Connected activities

How Fin Stats provide **info** on 3 **Types** of Activities

CRITICAL to understanding Financial Evaluation of Business Activities

Define Financial Statement- 3 BAs of a Firm – FS SFR GAAP

1.1.1. The Objectives of *ACCOUNTING*

Define Reporting Entity
Make available FI to stakeholder(s)
Current future investors
IFRS 3 specific investment decisions & 3 BA info need
Regulation reliance crucial
Accountability, Control, Decisions
Interesting & Socially Important

Raises cash from investors (equity/debt) to fund operating & investing activities

Long term investments a company undertakes to grow the business

includes production & sales of business goods & services & their supporting activities- CS,MS,M,HT

Fin Stat = Sig Info Source
Developed how: Assessed how: FR Accounting Rules
Assumptions and Concepts- Why?

Essential elements of FRA
Technical Aspects of FRA
Knowledge & Assessment
IFRS Analysis Reporting
FM Nature/ BAM Conversion/ RKA Tools

1.2. Basic Financial Statements

Standardized Summarization

IASB with IFRS FS Regulation

Local GAAPs (HGB) vs. Global IFRS

IFRS 4 < Annual Reports

Explanatory Footnotes + Executive insight

GAAP + IASB definitions

1.2.1. The Balance Sheet

- Lists Resources (Assets) and Claims (Liabilities/Obligations) against the Firm **at a Specific DATE**
- Represents the Cumulative Effect of all Prior Transactions and Events for *each listed element*

  Investing = non-owner Financing + owner financing

Accounting Equation

Assets: Measurable Resources of Economic Value Controlled by the Firm.
Expected to provide future economic benefits.
Assets allow the Firm to Operate and Earn Income

Liabilities: Measurable obligations arising from past events, often from acquiring resources, that require a future transfer of assets or services to other entities

Shareholder’s Equity: Measures Owners’ claim to assets of the firm
‘Net Assets’ debt hint
The sum of Capital the owners originally Invested & Net Income from business’ inception- Retained Earnings

2 Ways to Visualize/Understand Equation

- Sources of Financing Theory (Liabilities + Equity = A)
- What’s left Over view (E = Assets - Liabilities)

Conceptual Assumptions/Principles/Developmental Accounting

Balance Sheet *Balance Sheet E.g. (Copy/Enhance)*

Current Assets

Cash and Cash Equivalents- most liquid assets, immediate use, money in pocket or bank or near money same as cash

- NOT accounts receivables or stocks

- Cash: Checks not deposited
- Cash Equivalents (low risk of losing value with 3 months maturity):
    - Money market funds (pools money from investors for safe short-term cash-like securities) in 3 months
        - CDs- Certificates of Deposit
        - Repurchase Agreements (V short loans backed by securities)
        - Commercial Paper
        - T-Bills
    - Short term government bonds in 3 months

Marketable Securities- short term company held investments quickly convertible to cash in 1 yr.

- Stocks in 1 yr
- Bonds in 1 yr
- T-Bills in 1 year
- Money Market Funds in 1 yr

Non-Trade Receivables- Money owed outside of normal sales from the firm.

- Vendors who **owe** money for deposits/ rebates /refunds / promotions
    - ('Vendor non-trade receivable')
- Insurance claims
- Tax refunds
- Advances to employees (e.g. travel disclosure)

Net Account Receivables

Inventories/Supplies

Other Current Assets	Catch-all category for short term assets expected to be converted to cash in 1 yr- they don't fit other previous categories of current assets
	Too many items is a red flag for weak regulatory disclosure
	Miscellaneous convertible to cash in 1 yr, **prepaid expenses** (bringing asset value- rent/insurance), non-trade receivables, loans

Non-Current Assets

PP&E

Long Term Investments > 1yr.

- Investment in Associate- private company acquisition of 20-50% > 1yr. (else <20% is a financial investment here)
- Marketable Securities > 1 yr.

Intangible Assets

- Intellectual property, patents, trademarks, goodwill

Prepaid Expenses for more than 1 yr. (increases asset value)

Liquidity Check: Can company pay short-term bills with short-term assets? (Current Ratios).

Risk Insight: too much assets in long-term stuff might mean struggle with cash flow.

Transparency: It shows which resources are “cash soon” vs “cash someday.”

<u>Current Liabilities</u>

Unearned/Deferred Revenue

Taxes *Payable*

Short term Loans/ Commercial Papers, promissory notes, IOUs

Term Debt/ Current Piece of Long-Term Debt

Accounts *Payable*

Wages *Payable*

Other current liabilities

<u>Non-Current Liabilities</u>

Bonds Payable

Long Term Loan- 10 yr Bank/Mortgage

Lease Obligations

Long Term Provisions > 1yr.

Pension Fund Obligations

Other non-current Liabilities

- Deferred compensation
- Environment or legal provisions- cleanup costs/expected lawsuit
- Deferred tax liabilities
- Long term deferred revenue

<u>Non-Current Liabilities</u>

Common Stock and additional paid in capital; par value

Retained Earnings

Accumulated other Comprehensive Income/Loss

Balance Sheet E.g. (Copy/Enhance)

1.2.2. The Income Statement

'**Flow**' Statement focusing…

balance sheet changes affecting *net income*

But also shows **Gains & Losses** distinct from revenue & expenses

Period Performance + Formula

4 Definitions- customer, costs, asset liabilities, difference- NI loss

1.2.3. Statement of Cashflow

Net Cash Account Change in 3 Activities

Net Cash amount <u>spent</u> and <u>received</u>- operation investment & financing

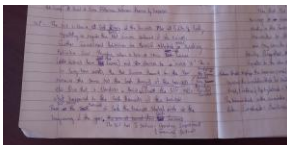
Net Income Difference

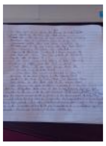
$Opening\ cash + CFO + CFI + CFF = Closing\ cash$

It is a measure of the overall cash balance of the business reflected in each 3 business activities

Does the business has good or bad cashflow?

Can it afford to keep going or cover its cash outflows?





1.2.4. Changes in Equity Statement

Components of equity, Changes therein + *OCI*

How the income statement links to the Balance Sheet.

Shareholder's Equity= Contributed Capital + Retained Earnings

AOCI

Contributed Capital= preferred stock + common stock + APIC

Total Value of Shares = Issued shares x Par value

APIC= (Issue Price-par value) x Issued shares

Retained Earnings= Net Income – declared Dividends paid

AOCI= cumulative income gains or <u>losses</u> Affecting equity but Not included in ***<u>net income</u>***

((Outstanding shares + Treasury shares) = Issued Shares) ≤ Authorized shares

1.2.5. Links Between Financial Statements

Fin Statements are for ‘Going Businesses’:
No intention of liquidating assets/terminating current trade agreements

Balance sheet- obligations and resources at specific date
SCF- beginning & ending Cash Accounts to balance sheet
Income Statement- beginning & ending *Retained Earnings to statement of equity changes*, Net Income focused
Shareholders equity- beginning & ending Equity to balance sheet

Investments: “Transactions that involve long-term or non-operating assets (or acquisition of any resources for use).”
If an inventory item will be used under the core operations of a business- operating activity, it’s not an investment activity.
IFRS says: classify by *what kind of activity it is for the business*

Liability means unpaid Expense
Expense- Item already paid for, or with little chance of being 100% returned

1.2.6. Jamie & Alex’s Coffee Shop- FS & BA of Small Businesses

Strategic decisions: company financing, assets management operating activities
Benchmarking industry performance

Going Concern

Define CPA

Define Accrual Book’s Definition

(financial consequences of transactions on the company’s economic resources or claims <u>in a specified period</u>)

“Clearly, absolute numbers are not enough to compare companies. Instead, numbers must be compared, something that ratio analysis help with”

1.2.7. <u>Which Business to Invest In?</u>

Quality of earnings

Depreciation: inventory assets/ accounts receivable
Accrual vulnerability/ future uncertainty

1.3. Key Ratios

Ratio defined

Analysis Part

Market Professionals Evaluates FS/ management/ equity & debt securities

Profitability & productivity: <u>*4 Ratios*</u>

1.3.1. ROA- return on assets

Return ROA- measures how much NI- net income management generated with average assets available in given year. <u>*Crucial*</u> acquisition & use <u>*Ability*</u>

Flow ROA= Net Income/ Average Assets

acknowledges Mean *change* of investments in 1 yr

Acquisition ROA

Profitability vs. Productivity

ROA= PM (profit margin) x AT (asset turnover)

Profit Margin PM= Net Income**/** Sales

Asset Turnover AT= Sales**/** Average Assets

Wong**.**

ROE= Net Income**/** Average Equity

<u>PM shows</u> how much part of your Net-Income comes from/ relates to <u>Total Sales</u>
The other part is expenses. Higher PM ratio = Profitability…

<u>AT</u>- how well management uses assets to drive sales
Measures how efficiently the business uses its assets

Wong 2017 says the <u>**profitability**</u> **component** measures the amount of profit from each dollar of sales while the <u>**productivity**</u> **component** reflects the effectiveness of generating sales from ASSETS.

ROE= (NI / Sales) x (Sales / Av. Asset) x (<u>*Av. Asset*</u>/ **Av. Equity**)

This disaggregation does more… it makes it ***possible to determine the catalyst for that specific change***… (Wong 2017).

:- Operational / asset / debt efficiency

If the ***ratio*** is ***calculated*** over an ***entire industry***, it becomes evident that ***some businesses*** have a ***very high PM*** because their ***product is unique***.

A ***product*** that ***differentiates itself*** from other products can be ***sold*** at a ***higher price***, generating more profit. **Other businesses** make a **profit** by **focusing** on **efficiency** rather than uniqueness

1.3.2. The DuPont Chart

What's a Du Pont chart?

Y= PM, X= AT, visualize ROA

FS Ratio Use Equally Important

Relationships (decisions & business picture)

DuPont Efficiency Line

Better AT **/** Higher PM- strategy choice

PM- differentiator/ innovator, fewer product sales higher-price

AT- Av. Asset turnover high because sales are high
PM lower because increasing profitability hurts sales

Healthcare vs Commodity stores

Financial ratios are *tools to analyse management & financial performance of a business*

Identifying **Relationships between** various **numbers** to generate a PICTURE of the **performance of a Business**

Accrual Income can impact the quality of company's ***financial statements*** & ***earnings***.

Some items aren't put on the balance sheet

Net Income and Cashflow are not the same.

Accounting regulated by standards but not exact science- uses professional judgment

General Accounting Principles

Characteristics, **G**oals & **U**sefulness of <u>IFRS Conceptual Framework</u>
How FS constructed from **Measures** of BA/ Financial Statement Effects Template
Period End Adjustments for Statements of Accounts
Similarities & Differences between Accounting Standards- IFRS, SMEs
Foundations of German GAAP + Evaluating Revolutionary Changes- **<u>ALMA 2009</u>**

2. General Accounting Principles

Introduction

Business owners want positive light

Thus, knowing how transaction activities impact FS

Appreciating the choices embedded in accounting rules

Communicating financial impact of business decisions to your network

2.1. Conceptual Framework Under IFRS

2 Conceptual Frameworks/ 2 characteristics, goals and usefulness

Who has standard Authority & Why?

IASB- OC/FR in CFFR
Goals CFFR

- **<u>Reliably</u>** Guides IASB to *develop* IFRS
- defines Accounting Concepts demanding companies to follow

IASB's objectives & concepts of Financial Reporting in their *<u>Conceptual Framework for Financial Reporting</u>*

2.1.1. CFFR Qualitative Characteristics

CFFR defines Useful FS Info.

Usefulness**= Relevant** + **Faithful Representation**
(Fundamental Qualitative characteristics)

Comparability, Verifiability, Timeliness & Understandability *also* ***Increases Usefulness***

Relevance= Predictive + Confirmatory Value
Ability to influence decisions
Materiality: increases degree of relevance

Faithfulness: Ideal Representation Impossible, Harrisson 2017
<u>representing</u> the occurrence of a <u>transaction</u>, or <u>event</u>
Legal contract basis + accounting economic basis
Faithfulness = <u>Completeness</u> + <u>Neutrality</u> + <u>Error Free</u> **EXPLAIN.**
(Aspirational approximation, impossible to list every single (opportunity installation cost))

Decision Usefulness Chart

Comparability-	Allows users to analyse FS in DETAIL. Only possible with consistent underlying preparation & presentation.
Verifiability-	Different knowledgeable & Independent Parties conclude specified info *Reported Faithfully*
Timeliness-	Info made available ASAP/ older info less useful
Understandability-	results from Classifying, characterizing & presented info *clearly* & *concisely*

2.1.1.1. IASB CFFR Constraints and Assumptions

User processing vs business preparing & reporting **Costs**

Accrual Accounting

Going Concern Assumption

2.1.2. Importance of the IFRS

Global standard

Interconnected Corps & MNCs- Trade, Int. money flows, production & supply chains

NAS Compliance Challenging National accounting standards

Company vs Investor

Cost, complexity, Risk.

2.1.2.1. Benefit Objectives of the IFRS

IFRS Goal- develop "standards bringing TAE to <u>World Finance Markets</u>
Serves public interest for Trust, Growth & long-term Financial Stability.

Transparency: because businesses report *similarly standardized* financial Info openly. Comparability and quality.

Accountability: **Information Asymmetry** (MD&A, conference calls, analyst reports)

Economic Efficiency: Reporting Costs, complexity & risk, faster business decisions & transactions

<u>IASC, IAS, IAS 41/ 17 IFRS</u>

2.2. IFRS for SMEs

FS 2 Step Preparation: Period Recording of transactions + Adjustments

Define Transactions- BA altering BS ALE accounts
Define Account
FSET Accounting Spreadsheet: same for large publicly traded company

2.2.1. Accounting and Balancing

Account Meaning

Category for tracking changes in 1 type of resource/obligation- a record summarizing all increases/decreases for that category

Explain FSET from scratch- 7321

Trial Balance & Transactions
NI to BS linkage
Separate line
Always 2 Transactions
BS Equation

either both sides ↑, both sides ↓, one side ⇅

Accrued **Expense**- costs already incurred, goods already used up, but no bill / invoice, short term liability, **cash NOT paid** yet, accrued interest.

Accounts payable- recorded when you are billed, not necessarily used up, short term liability, cash not paid off

Accrued **Debt**- debt, obligations incurred, cash not paid off

Accrued Revenue- customer not yet billed, but you already supplied goods or services

Accounts Receivable- customer billed, goods or services already supplied

2.2.2. Adjustments

Define Adjustment *__review__ meanings of 'accrual' in textbook*

Accrual accounting **separates** the 'recognition of economic activities' from 'when cash MOVES', and adjusting entries are a fundamental part of the accrual PROCESS.

Adjusting entries of diverse accounts ensures the proper recording of revenues when earned (accrued revenue and the earned portion of unearned revenue), expenses when assets used up or incurred (depreciation & prepaid expenses) and liabilities when incurred (accrued expenses) regardless of when cash is received or paid.

It uses accrual accounting to ensure the *timing of* revenue and expense *recognition* on the 'different affected 'accounts'' are accurately reflected/truly properly recorded.

So, these accounts are converted from a partial cash-accrual effects to a full accrual accounting basis at period-end

---

2 Questions AL overstated/ AL understated or unrecorded

Always 1 BS Account + 1 IS Account but not accrued expenses, priorly IS

Implicit vs Explicit Transactions Implicit have no immediate cash transaction *but Impacts Financial Account*

4+D Adjustments

Trial Balance Ending amount

4+D= DEPRECIATION, unearned **revenue**, accrued **revenue**, prepaid **expenses**, accrued **expenses**

2.3. Global- BillMoG and HGB in Germany

Majority *SMEs*– IFRS for SMEs not mandatory for **all** private SMEs

Publicly Accountable meaning

Lack Resources to comprehensively implement full IFRS

So IFRS for SMEs created

5 Simplification of Full IFRS

- Less Disclosures
- Irrelevant omission
- Simpler policy options, NOT others
- Streamline: RM principles
- Lay text Rewriting

< 85% of Full IFRS

2.3.1. Goals of IFRS for SMEs

Global standards- Insufficient GAAP, resources for Full IFRS

Global Recognition- ACCESS financial markets, compare, transparency

IPO Intermediary stepping stone

Tailored Simplification for SMEs

2.3.2. BillMoG and HGB

2002 IASB sets 2005 Public EU Comps., Germany 2003

HGB remains- statutory presentation & disclosure, tax, and profit distribution

Modern Legal Alignment- ALMA/ BilMoG 2008

German GAAP- creditor Tax info need, RM options, lower disclosure extent, and the prudence principle

Vs. IFRS decision usefulness

HGB vs IFRS/ startup expansion costs

Intangible, future benefits uncertain (market research etc.)

1. Period recording
2. Preliminary balance + BS Equation
3. Recording Period end Adjustments
4. Final Balance + BS Equation

Measuring Performance: Income Statement & Statement of Cash Flow

- chapter **OBJECTIVES**

3. Measuring Performance: Income Statement & Cash Flow

Introduction

CFO & NI = Financial Health & Performance
Differences, Similarities and Connections
Stock example
Earning Quality

3.1. Accrual Accounting

CFO- *Cash flow from Operations*

IS to CFO Relationship- Connections, differences & similarities

Accrual- Cash & non-cash effects

CFO SCF- Strictly Cash

Net Income vs Cash Flow **FIGURE**

- Accrual records revenues earned
- Accrual revenue **directly** matches exact expense/cost
- CFO Liabilities incurred not necessarily 'expensed'
- But ALL CFO liability costs **PAID** are readily 'expensed'

No *REASON* for NI Performance to *always* match Net CFO performance

Time + No Assets or Liabilities Except CASH = NI & CFO performance SAME

Figure Example Ending- Revenue & CFO inflow SAME
Expenses & CFO outflow SAME
NI & Net CFO Performance SAME

Accrual Superior but vulnerable to subjective judgments, earnings uncertainty & measurement errors

In period Accrual vs Out period CFO

3.2. Income Statement

Flow Summary of NI changes

3.2.1. Component of Income Statements, **A-G**

IAS 1 Table components stipulation

a) Revenue- insurance revenue
b) Other **C** Income/ Gains or Losses from amortized Asset Derecognition
c) Finance Cost
d) Income- Associate/Joint Venture via Equity method
e) **GL** from reclassified <u>financial</u> assets
f) Tax
g) Single A Discontinued Operations

Re Tax Finance, dis, disc. De.

Gains & Losses Separation

Nature- Expenses grouped by What they actually are, ignores purpose
No reallocation needed/ simpler/ Europe

Function Why spent/ Purpose/ role
Function allocation needed/ Detailed/ US
IAS Function **NEEDS** Additional Details

Company must pick one format, *not mix them*.

Test write nature and function **Sample**

Harrison 2017- history, industry nature of entity factors **/** *reliability* & relevance

---

Nature: Revenue, other income, WIP & FG from inventory changes, nature, depreciation & miscellaneous expenses, total expenses, Profit b4 Tax

Function: Revenue, COGS (+ selling costs), Gross Profit, Other Income, Admin delivery Depreciation R&D & marketing costs etc., Other Expenses, Operating Profit / EBIT, Interest/Finance costs, Tax, [Net Income]

3.3. Statement of Cash Flow

CASH period change
CFO/ CFI/ CFF
Define Cash & *Cash Equivalents*
Company *Checkbook* / *Cash Diary*
$Beginning\ BS\ Cash + Net\ Cash\ Flow = Ending\ BS\ Cash$

IAS 7 SCF primary SCF Accounting Standard
CFO Important performance Number & NI Complementary Competitor

Direct method: Analyses actual 'operating' cash inflow receipts and cash outflow payments to obtain net CFO independent of the NI statement. It is conceptually applicable to other SCF sections.

Indirect Method: Starts with NI, removes non-operating & non-cash items and then makes cumulative adjustments to NI to undo accrual effects.

3.3.1. Operating Activities Section

Method chosen depends on Current Situation

3.3.1.1. Direct Method

Used for fewer transactions with ability to inspect cash activity data
Strictly CASH moving transactions recognised
Operating, Investing or Financing
Inflow vs Outflow

CFO in: **deferred** revenue, converted AP, Revenue Cash; Cash not NI.
∴ depreciation & accrued interest part of NI but not CFO
CFO out: cash paid for operating costs and operating liabilities
CFI inflow & outflow; CFF inflow & outflow

Interpreting SCF- IMPORTANT
Investment and Life Cycle STAGE of Business

BETRAYS too much internal info- cash spent & received
Both methods **REFER** only to CFO but *direct method SCF applicable*.

3.3.1.2. Indirect Method- what to understand (probably?)

- Start with Net Income
- Add back non-cash expenses already included in NI
  - These do NOT impact cash outflow but reduce NI
  - Depreciation, stock-based compensation, impairment losses
  - 'True' non-cash items do NOT correspond to CFO. Non-cash Operating accruals move cash sooner or later in CFO and are resolved in step 4.
- Remove non-operating items included in NI
  - NI profit reflects Investing & Financing gains & losses
  - PP&E, Investment sold, and IFRS CFF linked *interest* cost on debt financing are typical examples.
  - So remove only their non-operating GAIN and LOSS included in Net Income.
- **Adjust** for changes in OPERATING working capital Accounts to convert accrual-based NI to Cash-based activity [**WC** adjustments]
  - Subtract increases in *Operating* current accounts (assets)
  - Add decreases in *Operating* current accounts (assets)
  - Add increases in *Operating* current liabilities
  - Subtract decreases in *Operating* current liabilities

Adjusting working capital to cash operating capital NOT about "what should be expensed."

They are about **UNDOING accrual accounting to get to CASH**. (How much cash actually flowed for that specific transaction)

An increase in [operating] current asset (that is not cash), aka FSET transfer [subtracting] from cash account to non-cash account, aka FSET buying a current asset with cash, doesn't impact Net Income.

Indirect CFO is TRANSACTION or PERIOD **specific**. Only the accrual effect for that FSET line or transaction is undone.

Every FSET line account change has meaning, but indirect method only cares whether that change caused a mismatch between NI and cash. The NI change effected is undone by 1 of the 4 categories of working capital adjustment above.

Items- a jet ski or manager, are linked to more than one lines- *more than one accrual effects*.

So another line on the same item- jet ski or manager is separately undone, sometimes in a different period not in the Income statement you have. You'll have 2 adjustments required. Everything eventually balances though. "Two accrual effects for **one** item"- jet ski, manager

Only the actual specific transaction given is adjusted- to reveal actual Cash Flow for that specific FSET line in that specific Period

An Exam or actual Income Statement may only list specified transactions. SCF only cares about actual cash flow for the NI transactions given in that period.

Transactions relate to an FSET BS account. You either use the working capital adjustment rules to adjust each transaction/ FSET line individually to derive net cash flow while IGNORING the Cumulative Account Sum or take the Cumulative Account Sum and do one adjustment deriving *account cash flow*. **You CAN'T do both individual & summary adjustment together.**

Every transaction or line adjustment on FSET is CUMULATIVE. The 'un-adjusted values' added up gives the SUM to be adjusted for that account in one go. A negative is a decrease and a positive is an increase.

So no worries, changes in each account is SUMMED up. Just remember the rules and apply it to that 1 account sum. That's all.

Adjusting sum totals literally undoes ALL line accruals *in Net Income* at once

OPERATING accounts NOT ***current*** accounts, current accounts NOT operating accounts. Current Accounts= Current Assets & Current Liabilities

Current OPERATING Accounts- A/R, accrued revenue, pre-payments, inventory,

Current accounts NOT operating Accounts- short term investments, PP&E, intangible assets

Current OPERATING Liability- A/Ps **ARE** an Operating liability.

Accrued debt= Accrued (wages, (utilities, rent & other operating expenses), *Interest financing costs*, taxes, dividends)

Operating accrued debt= operating wages & operating expenses

Adjust ONLY operating accrued debt

Accrued finance interest costs MUST be reported in NI. If so, necessary adjustment must be made. Under IFRS, interest paid may be classified as either CFO or CFF, *never CFI*. This classification choice affects only the cash flow statement, not Net Income.

CFF is not based on NI and does not require indirect adjustments tools. Hence, any accrual-related adjustment occurs in CFO, not in CFF. Thus, if classified as operating, it is sorted via WC accrual-related adjustment methods in CFO not in CFF.

The key ***CFO*** *question* is whether cash was actually paid- **whether NI ≠ cash flow**. So if financing interest costs are accrued- (expensed but) not yet paid in the prior period, WC adjustment calculations are required. Differently, when the accrued interest is actually paid in the latter period, no adjustment is needed because cash was actually paid. On this note be aware that paying accrued debt is the REVERSE of accruing an expense.

WC adjustments focus on operating items that affected NI without affecting cash, or cash without affecting NI. Care must be taken to not apply WC adjustment rules to EVERY transaction/ FSET line. Transactions may be '**cash-operating activity**' with no net income impact & cash moving later *but entirely ignored by WC adjustments* given no cash actually moved. Notice the incurring of **A/P** liabilities.

"Incurring a liability is prior to & different from… using said liability… or repaying it with cash." These are 3 different transactions/ FSET lines. Only the last two requires WC adjustment. Care must be taken to not necessarily see initial incurring of liabilities as a 'cash-**FLOW'** transaction.

Separate actual use of liabilities is 'expensed' in NI requiring an 'add decrease in current asset' adjustment to undo NI expensing. True net cash flow only results from actually PAYING the liability later- a cash-operating activity. [adjustment- subtract decrease of liability.]

You CAN'T do both individual & summary adjustment together.

Adjusting sum totals literally undoes ALL line accruals *in Net Income* at once

The focus of the Indirect method is not to reconcile individual cash-operating transactions. It is to reconcile cash-operating account TOTALS.

Adjustment ∴ applies when either, NI is affected but not cash, or cash but not NI.

FINALLY, Normal accounting systems only track ACCOUNT changes, including CASH. They do not inherently track movements by activity type- operating, investing or financing

By default BS account changes reflect cumulative effects of business activities. But it is not always one-to-one with a single activity type.

Changes in OPERATING BS accounts measure timing differences between NI recognition and Cash movement. This exactly what WC adjustment does.

NI is already fully computed under accrual accounting before SCF.

The indirect method reconciles net income to operating cash flow by adjusting only for those operating accruals that caused a mismatch between net income and cash. It uses period-to-period [Year to year] balance sheet changes & their matching OPERATING cash Flow ENDING Accounts as measuring tools rather than as individual transaction evidence.

SUMMARY- what experts remember (probably?)

Start with NI

Brief 3 Steps of the Indirect Method

It's not about 'what should be expensed' but 'UNDOING accrual effects to get to CASH'

All current accounts NOT cash-operating current accounts

Indirect CFO is Period/ Transaction SPECIFIC

1 Item but > 1 accrual effects

Individual vs Cumulative Adjustments- You CAN'T do both

Awareness & Acceptance of Incurring **A/P** liability false positive

Not all accrued debts WC adjustable, only Operating *accrued* debt

Finance Interest and the indirect method

<u>Accounting Systems and the indirect method</u>

3.4. Revenue Recognition Principle

Governs revenue recognition for ALL FS

Recorded *when* benefits can be *objectively measured* & earnings process substantially complete

Usually when service provided or goods delivered

Normal for cash received after revenue recorded

Complex Time Bundled

5 Steps/ **IFRS 15** Revenue from Contracts with Customers

- Identify customer contract
- Identify performance obligations
- Determine contract price
- Allocate price to each obligation
- Recognize revenue when obligation is Done

Parmalat scandal

NI = Performance, CFO = cash, *CFO* **critically important**, high cash or accruals = high risk.

Reporting & Analysing Assets: Balance Sheets

Understand accounting for receivables & allowance for BAD DEBT

Explain mechanics for *Inventory accounting* using Inventory Flow Assumptions

Discuss accounting for long-term assets + Calculating various Depreciation Methods

4. Reporting & Analysing Assets

Introduction

Reporting & analysing tangible and intangible resources/ 7 FSET

(cash, short term investments, A/R, Inventory, PP&E, IA, Prepayments)

New economy comps & Intangible Assets NEC Corp v Intel Corp

FSU Intangible Importance vs Accounting methods & issues

4.1. Defining an Asset

Criteria MUST

Current vs. Non-current assets

3 Chart definition control future past

IFRS CFFR definition

[Measurable] present economic resource [valuable item] controlled by the entity because of past events

Right with *ability* to *generate economic benefits*

also Resources used by business to generate profits

4.1.1. RECOGNITION

Criteria recording process

Recorded at initial cost

Carrying Amount/Value meaning

4.1.2. Cost of Acquisition

Monetary recognition & measurement types

Historical Cost: uses info from transaction price or *prior events*
Verifiability + Conservatism based Not opinion/market based

Historical update Exceptions: Financing interest, D/Amortization, Asset transfer/share, Impairment

Fair Value: Estimates True value of Asset, not purchase price
sometimes used under GAAP
Impartial E-valuation of Market exchange **Worth**

5 IFRS 13 FV Guidelines
Consistent UNIT of account
Well Ordered Market Main
Non-financial Asset HB Use
Applicable Valuation Technique from MP **Inputs**
General FV MP Procedures / Pricing

4.2. IAS 1 Inventory

No IAS 1 BS Prescribed listing order/ *Liquidity Choice*= relevance & reliability

Wong 2017 Liquidity

IAS 1 Current assets= most liquid/ quick turnover / time + liquidity factor

Expected to be consumed, realized (cash converted), sold in 1 OC/< 1yr whichever is shorter

Non-current assets= expected to be held/ benefit the business > 1 yr

Liabilities- dependent 1 yr settlement expectation

Working Capital Accounts

Working capital= current assets – current liabilities

4.2.1. Accounts Receivable

A/R Definition- *any* financial asset from *implicit*/explicit *contract*

Interest bearing receivables

Loan RP Example + Fin asset definition
fixed claim to receive fixed monetary sum

4.2.1.1. AR Bad Debt

Credit sales losses

Matching Principle- estimate losses, match revenue expenses

Not exact science (accounting measures)

Accrual Revenue requires reasonable estimate calculated & provisions made

New **Account** column: ADA / BDP

Bad debt Expense = Income Statement account

4 **Contra** assets: negative asset/ reduces asset account to reflect net asset balance

NOT a liability/ SCF

NRV meaning- *Amount from ordinary course of business net expectation*

doesn't change after writing off

Period Specific credit sale matching

Net **AR** book value vs. actual value/ (A/R - x) – (allowance - x)

Period Shifting Net Income violates accounting rules

NRV Now Zero 3

Investor/company reasons

Having no bad debt allowance NOT allowed under GAAP

4.2.2. Inventory

Inventory product business important/ service

Inventory Definition

Includes any <u>*purchased*</u> item ready to be sold, finished goods, works in progress and materials used for production process. **<u>Not long-term PP&E</u>**

COGS = Inventory

Usually Largest Operating Expense

Gross Profit

FSET Inventory

4.2.2.1. Inventory Cost IAS 2

Guiding cost concept for inventory same for assets?

Measurement of asset includes all costs <u>necessary</u> & <u>reasonable</u> to <u>place</u> asset in its current state of use

IAS 2- comprises all costs of *purchase*, *conversion* and costs incurred to bring inventory to its present *location* and *condition*

taxes, transport, handling, *other directly attributable costs*

Deductions- trade discounts, rebates, similar items

4.2.2.2. Lower of cost/ Market Test (for 1 batch) IAS 2

Inventories measured as lower <u>*of*</u> *cost* **<u>and</u>** *NRV*

Cost below market value

Investors future value Reassurance

IAS 2 NRV Period Write down- Loss/ Expense

4.2.2.3. Inventory Valuation & Reporting Income (Collective batches)

Net Income Significant impact

Changing costs requires deciding a valuation method for different Inventories

Core Accounting Principles- matching cost allocation over time

No one method Superior/ companies decide from several valuation methods

4.2.2.4. FIFO, LIFO and Average cost methods

Ending Inventory

FIFO

LIFO/ allowed by GAAP not by IFRS

Average Cost Method- yep, it checks out- Avr. Rev & everything

Companies pick a method aligning with particular financial interests

FIFO- F Reporting reflects sequential/ high net

LIFO- reporting likely reflects now market value ‘matching’ cost of last bought item

Average method- wants to make things simple by averaging out amounts

LIFO & FIFO Cash Accrual complete transaction cycle gives SAME RESULT

Sub-period breakdown betrays different results and approaches

RCGE Inventory Comparison **CHART**

4.3. Property, Plant & Equipment

Long term assets key assets in creating profits

IAS 16 PP&E are tangible items for use in the production of goods, rental to others or admin purposes

Expected to be used across several operating periods

Written up at the purchase price + incurred costs to set up for use

PP&E Set up Examples

4.3.1. Measurement after Recognition

IAS 16 Cost model vs Revaluation Model

Entire Class Choice

Cost **or** revaluation Reliable fair value Model **minus** Accumulated Depreciation

4.3.2. Depreciation

Depreciation questions how best to allocate the capitalized cost of the asset over its expected productive life as operating expenses

Process of assigning original costs to operating periods in pattern approximating how usefulness of asset is consumed

Consequence of the Matching Principle

Several operating revenues *Periods By Nature*

So Necessary to have logical & systematic way to distribute capitalized value cost against future revenue use

IMPAIRMENT vs depreciation writing down to recoverable value

Depreciation not valuation method

Expected future revenue uncertain

Does not necessarily equal real actual economic depreciation of asset

So unverifiable and potentially manipulable for earnings management goals

Why accountants consider nature & usage patterns before choosing D method

Residual Value- *after deducting cost of sale*

SLM when Equally Useful over life

DDB doubles SLM & **Special Case** so *doesn't* initially include salvage value

Final SYD 'COMPLETES' D value, not Salvage value, Unsuitable for high Ns…

Production instead of time-based method

S= $(\text{Cost} - \text{Salvage Value}) \;/ \text{Useful Life}$

DDB= $\frac{2\times 1}{N} \times (Cost - Accumulated\ Depreciation)$

factorial SYD= $(Cost - Salvage\ Value) \times \frac{Remaining\ Life\ of\ Asset}{\left[N \times \frac{(N+1)}{2}\right]}$

U= $\frac{(Cost - Salvage\ Value)}{Estimated\ Total\ Units} \times \boldsymbol{PERIOD}\ Units\ Produced$

Book Value

N vs **Remaining Life** vs **Accumulated Depreciation**

Depreciation Comparison Chart

4.4. Intangible Assets

Firms **SPEND** on acquisition, *development*, *maintenance* and *enhancement* of intangible resources

Intangible Assets- ***Identifiable*** non-monetary assets without physical substance

Resources whose Value based on representation not physical form

CHARACTERISTICS Not Financial Instruments

No physical form

Cost method SAME- Includes all amounts necessary & reasonable to place asset in intended useful state

Internally Generated vs Future benefits & legal fees

Purchased Assets

Intangible Chart attribute, type, purchased, internal, amortisation

Unidentifiable Intangibles cannot be identified or sold separately from the company. *Not a specific contractual right*. Reliable probable measurement

From Goodwill/ > fair value purchase, *no amortization*

Limited Life (Patents, Copyrights, Customer contracts, software, Licensed technology) *all with FINITE legal term*

Indefinites (trademark/brand name with indefinite cash flows & renewable protection)

Straight line Amortization

Testing intangible Impairment

scientific or technical knowledge
design/ implementation of new processes or systems
licenses
intellectual property

market knowledge
trademarks (brand names & publishing titles)
customer/ supplier relationships
customer loyalty
market share
marketing rights

4.4.1. *Research & Development*

IFRS allows Research vs Development Activities

RvD Chart- Definition, Accounting Treatment

Research- expense given future economic benefit uncertain

Development- possible asset given future economic benefit more certain

Accounting Intangibles ≠ Economic intangibles *≠ Firm Value*

Intangible book value understated/insignificant

Development + Legal Costs vs research expenses

feasibility, selling intention, ability to measure costs, probable future return, completion viability

4.4.2. Goodwill

Largest intangible asset in **BS** = only unidentifiable asset

Definitions Goodwill is the value paid above the fair market value of Identifiable Net Assets. Assets – Liabilities= Net Assets.
It is indeed value for the previously unmeasured, unreported, unrecorded assets/value obtained from the purchase of another entity.

Goodwill is Subjective- Explain IAS 36 yearly cashing units Tested

$$Price\ 2\ Book\ Ratio\ (P/B) = \frac{Market\ Valuation\ of\ Equity}{Book\ Value\ of\ Equity}$$

Goodwill Examples

reputation, market position, workforce quality, synergy, client/supplier relationships, internally created brand, future expected growth, Integration benefits, Unidentifiable intangibles

Net Asset @FV = Total assets fair value – Liabilities fair value
Goodwill = Entity purchase price – *NET ID Assets* fair value
Remember assets has both liabilities + equity, thus net asset different

Reporting & Analysing Liabilities & Equities: Balance Sheets

5. Reporting and Analysing Liabilities & Equity- Balance Sheet

Introduction

Sources of Financing FS/ Liabilities vs Equity
Debt Structured over Years = BS Money Time Valued
TV Fundamental to Financing & Accounting
Repay Structure How?
2 Equity Analysis

5.1. Defining Liabilities & Equity

Debt capital structure need

Notes + Bonds common huge funding

Nature Comparison

Debt Cheaper?

Return function of risk

IFRS CFFR Liability requirements/ 3 Obligations *past transfer*

IAS 37 PL difference= fixed amount, deadline [no mention of interest]

Present obligations with uncertainty *categorized under liability*

5.1.1. Current Liabilities

Current liabilities in BS *typically Operating Accounts*

Face Value NRV Liabilities (no change in FV to be paid)

IAS 1 Liability Requirements- defer trade 12/OC

Cash paid after- goods received (liability) vs goods consumed (Accrued Debt) vs loan repaid over agreed time (Long Term Debt)

Operating or Not?

Current/ financing liabilities ≠ Operating liabilities

A/P operating Liability

Accrued Liability Examples

5.1.2. Non-Current Liabilities- LTD

LTD reported in BS at present Economic value of Future Value *think inflation etc.*

3R Patterns + Amortization Repayment Table

Final Lump sum vs ***Instalments*** vs Coupon Annuity Interest Pattern

Amortization R Table: E Dates, Cash Flow, Interest, Book Changes in Balance, Balance Owed, Total

Interest due *on due date*- year end

Different effects on Cash Flow, & Interest Payable (NCF)

2nd pattern fastest repayment + Lowest Interest

5.1.3. Analysing a Company's Use of Leverage

Before TV Money → Risks & Opportunities ***LTD*** → **High vs low Leverage**

Debt supercharges the usefulness of equity

Though debt costs & means less total profit compared to equity, that is still valuable additional profit from debt, separate from equity

Hi/Lo C Table: ROA, EBIT, IE, EBT, Tax, Net Profit, ROE vs. ROCE

Hi- higher **IE**, comparatively less **P**rofitb4Tax, lower tax

Lo- bad Tax, Good PROFIT, lesser IE

39% of INTEREST is TAX Deductible, so ATCD tied to 61%

$ATCD = r_d \times (1 - Tax\ Rate)$ $= 12\%\ x\ (1\ -\ 0.39)$ = **7.32%**

It ∴ actually costs **y%** *NOT* **x%** to finance…

You pay Full Interest with less tax / 7.32% Interest because tax deduction

~~Cash paid $48,000 tax shield write off $18,720 ATCD $29,280~~

~~($400,000 x 12%) x (1 - 0.39) = $29,280~~

7.32% *of Principal* = 61% of IE paid = $29,280

ROA ATCD comparison Works, Asset > debt, loan affordable?

ROA *ignores* financing, *not tax*… = ***EBIT (1 – tax rate) / Asset= 8.54%***

Standard ROA vs Operating ROA- wants to show return on assets intentionally ignoring how any firm was financed (debt vs. equity).

ROA vs ATCD Comparison Factors:

- ROA- Tax, ATCD- IE & r_d tax rate

ROE Comparison

bad accurate for hi leverage better for lo leverage
Hi ROCE includes majority profit from liability vs Lo ROCE

So net income can be lesser due to interest expense even though debt financing being "cheaper" than tax rates and reducing taxable income

Debt helps owners earn more per dollar of equity, not necessarily more total dollars. But higher debt means more income from debt not equity

Bad Earnings Year= Less EBIT

Less profit to pay IE

Vs. higher profit for low leverage company

ROE *negatively* affected here

ROA likely < ATCD = DANGER

Companies MUST still pay Creditors in Bad Yr.

5.1.4. Equity- Book value not Market Value + SCF + BS Equation

5.2. Accounting for Debt Financing- be patient, go line by line

Research Bonds. Explain Bonds.

Present economic **V**alue of a matching **F**uture **V**alue

FV *of* ***Any Asset*** *linked to* ***PV*** *by* ***D****iscount* ***R****ate*

$$Present\ Value\ of\ future\ cash\ flows = \ Promised\ future\ cash\ flows/Discount\ Rate$$

PV x DR = FV, PV = FV / DR, PV = 100 / **(1+0.8)^1**, (1+0.8) ^1= Discount Rate

^1- power index matches different amount of time into future (intervals) at which each future value payment is made, ^1 in respective formula raises PV or drops FV

Bond Interest rate payments & FV Principal **vs** their discounted Present Value

Fixed and issued prior for future collection **vs**. market Fluctuation (macroeconomic, supply, demand & buyer)

Bond Interest pay ***rate*** of FV Principal ***Calculates*** *FV Interest coupon* ***AMOUNTS*** *paid*

Interest coupon ***rate*** of FV Principal ***not*** PV Discount Rate

Discount rate/ market yield/ market effective rate
set by market forces, makes use of simple & compound interest math

$$Bond\ price = PV\ of\ All\ future\ coupon\ payments\ + PV\ of\ the\ principal$$

Bond holders/buyers invest today's principal and interest but gets tantamount future economic value of their investment

Discount rate *formula above* calculates PV of each Interest Coupon paid overtime **individually** **(^1)**

Riskier asset, higher DR, higher Coupon return, lower investment

Premium vs Discount vs Neutral bond price

Business decisions necessary= **Balance Sheet Reporting**

Bond Pricing Chart: Principal, Annuity Coupon, BS Total, FV Bond, PV

Same Idea of prior formula, **USED to <u>simultaneously</u> discount <u>ALL Coupon Amounts</u>**

$$Annuity\ Coupons = \left(\frac{1 - \ (1 + 0.04)^{-20}}{0.04}\right) \times FV\ Coupon\ Amount$$

FSET **Bond Price Repayment**- Discount vs Premium Amortization

<u>Bond Price received increases **Cash** & Liability (LTD) columns</u>

<u>Bond *price*</u> liability written under **LTD column** for *Later repayment of **<u>bond principal</u>***

Coupon Interest Payments 'expensed' written only under **Equity column** overtime

FSET **Bond Coupon Interest**

Issue Discount: - (Equity) Interest Expense + Liability = - Cash

Issue Premium: - Interest Expense – Liability = - Cash

Buy Discount: +Revenue Interest Recognised = +Bond Investment +Cash

Buy Premium: +Interest revenue recognised – Bond Investment = + Cash

Coupon rate affects bond liability investment price

Coupon rate < market rate

Coupon rate > market rate

Coupon rate < market rate

Coupon rate > market rate

But *premium bond price* written under liability > FV bond principal to be repaid

Gapping Extra-Premium Amortization amount *individually deducted* to complement *each Coupon **'expensed'***, bond price deducted over time <u>until = FV bond principal</u>

$$(\boldsymbol{Cash})\ Coupon\ payment = \ Premium\ deduction\ (\boldsymbol{LTD})\ +\ Coupon\ expensed\ (\boldsymbol{Equity})$$

$$Coupon\ expensed/'\boldsymbol{Interest\ payment}' = Current\ Bond\ Price\ x\ (Discount\ rate\ -\ 1)$$

$$Coupon\ expensed\ (\boldsymbol{Equity}) - Full\ Coupon\ payment = \ Premium\ Amortization\ (\boldsymbol{LTD})$$

Note: Current bond <u>Price</u> constantly changes due to the previous premium deduction

FV bond principal repaid and deducted under LTD column at maturity

Contrarily, *discount bond price* < FV bond principal to be repaid

Bond Price received increases **Cash** & liability columns

LTD Column records= FV Bond Principal & Contra-liability Discount

$$Contra\text{-}liability\ Discount = PV\ Bond\ price - FV\ Bond\ principal$$

Contra-liability Discount cancelled out to match FV bond principal over time as each coupon interest is paid- by *adding* discount amortization *deducted from* Equity.

FSET always balances out, liability principal value 'short' deducted from equity, cash coupons paid deducted from equity

Note: Coupon expensed payment written only under equity columns

But *Discount coupon* expensed *more than* coupon payment.

This Extra Equity coupon 'deducted' and 'added' to liability column

Missing contra-liability *discount amortization* completes FV bond principal value

$$Coupon\ expensed\ (\boldsymbol{Equity}) - (\boldsymbol{Cash})\ Coupon\ payment = Discount\ Amortization\ (\boldsymbol{LTD})$$

Premium coupon expensed payment *less than* coupon payment

Research Tip: ask AI on FSET accounting for '**buying'** a bond, *note PV of debt concept used on Balance Sheet for 3 Loan Repayment options* (pattern 1= lump sum PV, pattern 2= PV of instalments, pattern 3= bond coupon style PV)

5.3. Accounting for Contributed and Earned Capital

CC vs EC

CC not necessarily good

Companies earn through profitable operations

Retained earnings Reinvested

5.3.1. CC= Stock accounts= Par value & APIC

5.3.1.1. Common Stock

Typical & usually voting stock

Owner decision making

3D Features

- VR: Most Important share class- decides board of Directors
- Proportionate Pre-emptive Rights
  - Good/bad dilution, MAP BEX
  - Book value of CC- proportional shares, Equal buying, minimum fair value price, anti-dilution extra shares, constitutional issue approval, post dividend issue
- Dividends

5.3.1.2. Preferred Stock

Special Rights

Fixed *cumulative* dividends

Usually without voting rights

Features Variable

5.3.1.3. Par Value and Share Premium CEP (APIC)

Balance Sheet Report How?= $Nominal\ par\ value + APIC$

Par value IDs #of shares issued in firm's owner capital

& so Relative ownership power

APIC above Par in separate APIC account

Market trading price vs. company issue price

5.3.1.4. BS Presents CC

IAS 1 Info for each Category

- Authorized #
- Issued #
- 1 Par Value
- Outstanding Reconciliation #

$Book\ Value = Total\ par\ value + Total\ APIC$

5.3.2. Earned Capital = LIFE Cumulative NI – Declared dividends Paid

5.3.2.1. Reserve Accounts

For protection from losses or set aside for specific purposes

Non-distributable & retained Pending decisions + statute law

3 Primary Transaction Accounts (Par, APIC, Retained)

5 reserves (legal, revaluation, reserve for owned shares, general) vs AOCI reserves

5.3.2.2. Retained Earnings

ChatGPT's IFRS Figure 15 Retained earnings

AOCI Reserves- Complete measure of all changes in Equity from all non-owner activities

Pension **L**iability adjustments, exchange rates, unrealized IL Sec+ Deriv

<u>IFRS SOCIE- Transactions Vertical + Accounts Horizontal + Total Equity</u>

Opening Equity │ [Period End] │ *Adjusted* Opening │ NI │ OCI + Total Items │ Share Issues │ repurchases │ REISSUES │ Dividends │ Reserves │ Closing Equity

5.3.3. Share Repurchases/Treasury Reduction

Companies repurchase shares

Gives Cash invested back 2 stockholders like dividends

<u>Separation of Common & Preferred Stock</u>/ Reserve Accounts:

- Common Par Value
- Preferred Par Value
- APIC Common stock
- APIC Preferred stock
- *Treasury Common stock @Repurchase Price*
- *Treasury Preferred stock @Repurchase Price*

<u>APIC is a cumulative aggregate account</u>- there actually **isn't** a uniform APIC per share value

A repurchase *reverses/reduces* equity value invested priorly made by shareholder.

- It is a negative equity item / contra-equity
- It represents equity removed from circulation (actual economic sacrifice made by company)
- Does not reverse authorized issuance But
- Reverses the issuance of outstanding shares
- The investment amount sacrificed (deducted) is recorded in the contra-equity account 'triggering' the same 'reserve for own shares account' amount

$$((Outstanding\ shares\ +\ Treasury\ shares)\ =\ Issued\ Shares)\ \leq Authorized\ shares$$

FSET= Cash column ↓, Negative Treasury contra *Equity* ↓

Repurchase= No written recognition change to par value

Repurchase= No written recognition change to APIC

Repurchase= Only written recognition via Treasury contra Equity ↓

Stock transactions (& Treasury transactions) never create gains or losses in net income. Handled ONLY within equity.

Treasury is a *temporary holding account*, not reserve for treasury gains/losses

Repurchase Treasury price represents the ***loss value*** of investment (added up/aggregated for each repurchased share)

Individual repurchase price *tied to* an individual share

Any Reissuing treasury shares may finally cause gains/loss **in APIC**

Repurchase ignores profit & losses, it wants capital returned and treasury shares to become outstanding before recording profit or loss in APIC

FSET Reissue= Cash column ↑, cancels Treasury contra *Equity* ↑

Any Reissue of treasury shares '*removes'* (cancels) relevant share values from treasury account

Reserve for own shares 'value' paid 'restrict' retained earnings. *Reissuing* treasury shares cancels contra equity & 'releases' reserve for own shares value paid to retained earnings- equity

Reissuing treasury shares ***@Repurchase Price*** *restores par value & APIC investment* for relevant shares
(Cash column ↑ cancels Treasury contra Equity ↑)

Reissuing **below repurchase price** maintains temporary partial loss value per share **so *APIC deduction necessary***, if insufficient **partial loss** negated from Retained Earnings
(Cash column ↑ still entirely cancels Treasury contra Equity ↑)

Reissuing above repurchase price creates *additional APIC gains* under **relevant** APIC Account Common/ Preferred APIC account
(Cash column ↑= cancels Treasury contra Equity ↑ Additional APIC ↑)

Par value and APIC **economically** spent/invested long ago (to Buy assets, Fund operations, Invest, or Repay debts)
But ***still legally*** represent stockholders ownership claim/investment on firm value.
Such claims generally **cannot be distributed** unless appropriate, only dividends generated on stock invested can be distributed

6. Financial Statement Analysis

Introduction

Details of How Decision Makers using FS Info

Data Reported not directly useable in decision making

Data need to be analysed, compared, decoded and interpreted

Techniques, ratios & other methods for FS Analysis/ Investor Creditors

Understanding 'what goes into FS' key prerequisite for good analysis & NOT Separate Endeavour

Unit Maps closely Analyst, consultant and Investor careers

6.1. Horizontal and Vertical Ration Analysis

Investor understanding Characteristics of future revenue & evolution of past NI over time

Horizontal- evolution of a specific item over time

Trend- presents % change against like value from a base year

$\left({}^{Year\ 2}/_{Year\ 1}\right) - 1$

6.1.1. Usefulness & Limitations- Horizontal Analysis

Short + Long Term Trends

Useful for immediate BM changes

Long term analysis *not always Meaningful*

Difficult Differentiate- Intentional Changes vs Int. External **EC** Forces

Meaningful= Business Model & Environment Stable

6.1.2. Vertical Analysis

Vertical **Common-Size** Analysis

Measurement of an item **as %** of a reference item in same FS

Expresses ALL FS Numbers **as %** in such way they sum to 100%

Provides *In-Period understanding* of *relationship between Accounts*

How FS structure has changed

6.1.2.1. Usefulness & Limitations- Vertical Analysis

Reduces problem of uncommon size differences

Helps analyst formulate hypotheses on Best & Sustainable BM- Why best BM

Effective for Same Industries

But limited challenging comparison from Different FS Structure, Presentation & Accounting Approaches

Table 39- Horizontal vs Vertical Analysis

ROS = PM profit margin

6.1.3. Ratio Analysis

Ratios 'abstract' (reveal) size effects of data

They allow both small & big firms to be compared

They *assume* a *meaningful relationship* between *2 elements*

Past **How** & Forecast How Much?

Widely used

Caution- ratio definitions Not Standardized

Categories

- Profitability Evaluation & Generation
- Short- & long-term Liquidity Evaluation
- Evaluation, Solvency & Financial leverage
- Shareholder returns & firm valuation Measurements

Basis: No single accounting # sufficient analysis indicator

3 Factors of Good performance **#**

- More CC invested, more profits
- Total size of Company/ Assets/ *ratios*
- Type of Analysis/ Question- different users, different metric details

Industry + Time comparison.

6.2. Analysing Profitability, Liquidity and Solvency

6.2.1. Shareholder profitability

ROE- *shareholder*, return on average equity

Dividend payout Ratio $= \quad Dividends/Net\ Income$
(getting the most dividends from NI)

GP Percentage Ratio- $= \frac{Gross\ Profit}{Total\ Sales}$, Markup profit %, **PM** compared

6.2.2. Creditor liquidity

$$\boldsymbol{WC\ Current\ Ratio} = \frac{Current\ Assets}{Current\ Liabilities}$$

Measures liquidity, > means creditor confidence, extra liquidity

$$\boldsymbol{Acid\ Test\ Quick\ Ratio} = \frac{Net\ A/Receivable + Marketable\ Securities + Cash}{Current\ Liabilities}$$

Can current L be paid with easy cash? - more *conservative* liquidity measure

< Current ratio and often <1

Excludes Prepayments and inventory

6.2.3. Solvency

$$Debt\ 2\ Asset\ Ratio = \frac{Interest\ bearing\ debt}{Total\ Assets}$$

$$Debt\ 2\ Equity\ Ratio = \frac{Interest\ bearing\ debt}{Common\ shareholders'\,equity}$$

Usually only *Interest-bearing debt*
Loans & bonds (principal owed) NOT a/p, accrued D, UR, provisions or taxes

Important <u>measure of valuable risk</u>

Relative Importance of finance **sources**

Risk not intrinsically good/bad, investor POV useful
Which ratio measures are horizontal, and which are vertical?

6.3. Using Accounting Information in Valuation

IFRS compliance

Both P/E & EPS common for valuation but who says dividends per share?

Flow EPS most widely cited and PAM

Analyst bottom line measure because

$$NI\ available\ to\ \boldsymbol{common}\ shareholders = NI - Preferred\ stock\ dividends$$

$$Basic\ EPS = \frac{Net\ income - Preferred\ Stock\ Dividends}{Weighted\ Average\ of\ Outstanding\ common\ Shares}$$

$$\mathbf{1}\ Weighted\ Average = Period\ Share\ Outstanding \times \frac{Time\ Outstanding}{Total\ Time\ (12\ months\ or\ 365\ days)}$$

Convertible preferred shares= Complex capital structure, share uncertainty

EPS <u>adjusted</u> for <u>other potential dilutive securities</u>

$$Diluted\ EPS = \frac{Net\ Income}{Weighted\ Average\ of\ Common\ Shares + Common\ Stock\ \boldsymbol{Equivalents}\ (Preferred\ Converted)}$$

IFRS required in <u>SCI</u>- stockholders' *Knowledge* & Awareness of potential dilution

$$Price\ to\ Earnings\ Ratio = \frac{Current\ Stock\ Price}{EPS}$$

Stock price current as <u>*multiple of **Earnings**PS current*</u>- assume DEPS unless otherwise

Difficult to interpret
- Some analysts say high PE= more future earnings
- Others, too costly, unprofitable
- EPS changes every year, price every day

P/E- Is it worth it? **No substantive evidence.**

6.3.1. Future Valuation

Popular simplicity- use only as a rule of thumb

But Important stock valuation help to analysts

EPS & P/E Valuation Matrix figure 16

- Current EPS and P/E ratio
- **Next yrs.** sell side Analyst Hi-Lo EPS / personal (industry & past trends)
- Alpha beta competitors Hi/Lo P/E
- Extrapolate *forecast* value *types*

P/E provides useful framework for assessing *relationship* between *accounting earnings* per share unit and stock price *market valuation*

7. Accounting Illustrated- A case study

Introduction

Provide Technical skills for Systematic Interpretation of Accounting Information

Interpreting Nuances in FS reporting

Fundamental Investor- stock Price vs true Value

FS from *non-speculative* accounting rules + Audited

Annual Report- Interpreted how, right Analysis approach

FS Multi-national Analysis

7.1. Application of Accounting Principles

Father of Value Investing

Criteria to **Uncover** value in a stock

1st necessary to Understand Opportunities & Risk of firm's BM

4 FS NOT full disclosure

Unreliable Quantification

Annual Report Verbal, Qualitative & MD&A examples

Important part of firm's Info environment

7.1.1. Business Strategy

Industry & specific Line of Business

Potential investors research strategy, profit Sustainability and risks

Data in annual Report- **10K form** explained

7.1.2. BMS Risks & Opportunities

Licensing

Size

Distribution System & Independent bottlers

WIDE range of products

200 countries

7.1.2.1. Risks

Affects Financial Results

Accounting rules

Laws & regulations

Debt interest rates

Foreign exchange rates

47 Same as competitors

7.1.2.2. Opportunities

Business Objectives

Strategic New Business acquisitions/ Alliances

Healthier options

Developing world

Ready2drink coffee

7.1.2.3. Porter's 5 Forces

List forces

All forces IMPORTANT to strategic Planners & Investors

<u>New Entrant Threat</u>

- Differentiation, giant economies of scale, critical capital requirements

<u>Supplier Bargain Power</u>

- Standardized ingredients
- Easy switching
- Large customer
- Strong power

<u>Customer Bargain Power</u>

- Quality loyalty, repeat price purchase

<u>Substitute Threat</u>

- Price, quality, reputation, differentiation, health

<u>Current Industry</u>

- Few big Competition- large market shares
- Every Strategic Move visible

7.2. Analysis of Accounting Information

Does *accounting* reflects business REALITY

Du Pont analysis helps CRITIQUE Industry performance,
<u>FS reality</u> *already non-speculative*

Across both Time & Industry

7.2.1.1. Du Pont Model

Is this business reality worth investing?

Increasing ROE

Compare industry

EXPLAIN

AT- Underlying asset use Dynamic *Important*

Total AVERAGE Leverage- NOT Debt 2 Equity, = *Remaining Debt Payable*

PM- NI from sales *after tax*

ROE- may Increase from ↑ PM & Leverage use

ROA- *profit as a portion of assets*, may decrease from low PM

7.2.1.2. ***SUMMARY* + *Du Pont Ratios Competition Comparison Table***

7.2.1.3. Graham's Criteria

Cheapness & Quality factors

2-part analysis- high earnings dividend returns, low stock price

Stock market data in Yahoo Finance under Historical Data

NYSE

Debt 2 equity < 1
Total Debt < Net Current Capital x 2
Current Working Ratio ≥ 2
No more than 2 declining NI years >5% in past 10yrs
Yearly EPS growth in past 10yrs >7%

EPR ≥ AAA bond yield 2x
P/E < 40% the *Average* 5yr P/E
Dividends > 2/3 of AAA Corporate bond yield
Stock Price < 2/3 Tangible BVPS
Stock price < 2/3 NET current asset / Share

7.2.1.4. Relationship between Cash Flow & Net Income

Predicting future Earnings valuable and important

Evaluating NI CFO movement over last years

Assessing EQ= True Reality

EQ= NI – CFO

EQ= Correlated CFO & NI

EQ Insight CHART

7.3. Recommendations Based on Accounting Information

True Intrinsic Value & forecasting same?

Forecasting= listen Most recent Conference Call/ quarterly financial results, Q&A

Get recent analyst report, use to forecast- earnings, growth & price targets

Note: Topic beyond scope but Benchmarking Analysis to Investment Professionals & the Market Important for Fundamental analysis

Your estimated intrinsic valuation

Do Sensitivity analysis- vary EPS and multiples applied

3 Comparison- Intrinsic vs Market vs Current price

$Deviation\,\% = ([Market\ Price - Intrinsic\ Value]/Market\ Price) \times 100$

7.3.1. Investment Recommendation

Strong buy, buy, hold, sell, strong sell

News not accounting- other Impacting factors not from accounting

3 steps of fundamental value investing

- Business Strategy risks & opportunities
- Accounting Analysis via Industry and Time Comparison
- Financial Analysis via forecasting Info to value firm